

Automatic storage tiering

Automatic storage tiering, or auto tiering, is built on the concept of Hierarchical Storage Management. The dynamic movement of information between disks of different types to meet the performance, cost and capacity is auto tiering. The movement from one level to another is triggered based on policies set by the storage administrator.

Multi-tenancy

Multi-tenancy is about ensuring that data that is being accessed by an application has a boundary that allows access to only the applications that have permissions. This will enable the same storage to be used by different applications within a single enterprise or different enterprises, without compromising on the data security. Multi-tenancy has the most utility value when used in the cloud environment, where the service provider hosts data services for different companies in the same industry.

Future trends

Two prominent trends that will emerge in 2012 are the adoption of SSD/flash as part of the Tier-1 storage layer

and increased storage utilisation. Another trend that could go beyond 2012 is the convergence of servers and storage devices into single systems. **END** 

References

- [1] Storage and storage virtualisation tutorials can be found at: http://www.snia.org/education/storage_networking_primer
- [2] Storage Virtualization: Technologies for Simplifying Data Storage and Management by Tom Clark
- [3] All about LVM - <http://www.markus-gattoi.name/ws/vm.html>
- [4] A Quick Guide to iSCSI on Linux - <http://www.cuddletech.com/articles/iscsi/index.html>
- [5] Open iSCSI implementation - <http://www.open-iscsi.org/>

By: Janardan Revuru

The author is currently working for the Enterprise Storage, Servers and Networking division of Hewlett Packard, Bengaluru. He has more than 15 years of experience working for Siemens, Motorola and Avaya. He is fond of open source software and Web applications, and can be reached at Janardan.Revuru@gmail.com.

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Linux-based systems' migration is quite seamless—everything just works. However, it's not that perfect for Windows-based systems (see Figure 9); they require additional tweaks to replace VMware-specific drivers with standard Microsoft drivers. After that, you can migrate the VMs into Proxmox. Fetch the 'mergeide.reg' file from the Proxmox site, uninstall the VMware Tools package, and import the registry file while the system is still running on VMware:

```
Start >Run Cmd >cmd >start mergeide.reg
```

Now restart the Windows VM, so Windows installs the standard drivers. Power off the VM in VMware Server. Create an empty VM as 'Fully virtualised (KVM)' on the Proxmox host. Copy the VM image from VMware Server to Proxmox, and power on the migrated VM. Windows will detect new devices, like the video and network controller, as well as new ACPI devices, and install drivers for them. Then assign an IP address and that's all. You've successfully migrated the Windows VM!

Direct VNCViewer access

You can access any VM without a Web browser and JVM. Use the following VNC redirector at the Proxmox host side:

```
# nc -l -p 5900 -c "/usr/sbin/qm vncproxy VMID PASSWORD"
```

Connect a standalone vncviewer to the Proxmox host, port 5900. This will redirect you to the necessary VM. This trick could be a real help, especially if you've incorrectly set up network configuration inside the VM.

A lot of powerful open source virtualisation products have appeared and have been in active development over the last few years. Proxmox is one of them. One of its big advantages is that it allows you to deploy a virtual environment from scratch very quickly, and painlessly migrate VMs from different virtualisation products. The uses of Proxmox are quite wide-ranging—be it a fault-tolerant Web hosting platform, a multi-purpose compilation farm for developers, and so forth.

Unfortunately, such interesting things as how to create a cluster from two Proxmox servers, adding iSCSI storage pool, etc. are beyond the scope of this article. But I'm sure that you're already convinced that Proxmox is a powerful, simple and elegant solution for your future virtualisation infrastructure! **END** 

Links

- [1] http://en.wikipedia.org/wiki/Kernel-based_Virtual_Machine
- [2] <http://en.wikipedia.org/wiki/OpenVZ>
- [3] http://en.wikipedia.org/wiki/Occam's_razor
- [4] <http://en.wikipedia.org/wiki/802.3ad>
- [5] http://pve.proxmox.com/wiki/Get_Virtual_Appliances
- [6] <http://www.proxmox.com/downloads/proxmox-ve/misc/59-mergeide>

By: Anton Borisov

The author has specialised in Linux and FOSS technologies for more than a decade. His professional spheres of interest include, but are not limited to, robotics, embedded systems, statistics and algorithmic methods.